

Project Title: Restaurant Review Sentiment Analysis System

Intended Use of the System

The primary users of the system are restaurant owners/managers and potential customers.

The Owners/Managers will be interfacing with the system through dashboards which consolidate all kinds of customer reviews and feedback available online. This way, Managers can detect trends, common concerns and areas of improvement without having to slog through thousands of reviews.

Diners will get a simple way of gleaning key insights about the restaurant in question to decide where to eat.

Overall Functionality

The system is designed to address information overload in restaurant reviews. At present, customers, as well as management, find it difficult to sift through heavy textual content stretched in multiple platforms like Google or Yelp or social media. The system shall:

- bestow review data from various platforms;
- preprocess/clean text for uniformity;
- classify positive, negative, and neutral sentiments;
- identify subjects-aspects (e.g., service, food quality, ambiance); and
- display results through the dashboard, charts, and word clouds.

The system shall perform text-preprocessing using various NLP techniques. This shall be followed by sentiment detection through classification using ML classifiers like the Naive Bayes or Transformer-based models, followed further by trend analysis for ever-growing customer insight.

Main Components of the System

1. **Input and Perception Module:** Captures raw reviews from external platforms or datasets and pre-processes them (tokenizing, removing stop words). Thus, for input, it accepts various raw reviews which indeed are drawn from an external ambiance or perhaps the different dataset maintained at the processing site.
2. **Knowledge Base:** Maintains cleaned reviews, sentiment labels, keywords, and topics authoritatively extracted.
3. **Inference/Reasoning Engine:** Run ML/NLP-based models of sentiment classification and trend detection.
4. **Learning Module** consists of continual training of models to stay updated with ever-changing language and slang.
5. **User Interface:** User Interface: There are the following two versions implemented: the interactive screen with the charts, word clouds, and filtering options for managers, whereas diners are here provided with the rather plain and concise view.

System Requirements Document (SRD)

1. Overview of Functional Requirements

The system is to summarize sentiment and trends in customer reviews of restaurants. The intended users of the system are:

- Management/Owners: View sentiment analysis for actions on service/menu;
- Diners: View what is being said.

The system will perform AI functions such as:

- NLP preprocessing of textual data;

- Sentiment classification (positive/negative/neutral);
- Topic extraction and trend analysis; and
- Adaptive modelling over new data.

2. User Stories

- As a restaurant manager, I want to see daily sentiment summaries an urgent service-related matter might be sorted out."

Pre-conditions: The reviews are gathered and processed.

Post-condition: A sentiment breakdown is displayed in the dashboard.

Type: Simple

- As a diner, I want to see a restaurant's sentiment highlights so that I can decide what restaurant to choose for dinner.

Pre-conditions: There is restaurant review data available.

Post-conditions: The customer-side view displays a summary and word cloud.

Type: Simple

- As a manager, I want to see the analysis of sentiment based on trends over time in order to measure the implications of operational changes.

Pre-conditions: Historical review data are available.

Post-conditions: Time-series charts show the evolution of sentiments.

Type: Epic

- As a system administrator, I want the AI model to automatically update with new data in order to ensure the system's accuracy over time.

Pre-conditions: New labelled reviews exist.

Post-conditions: The model is trained further and becomes more precise.

Type: Complex.

3. Non-Functional Requirements

- Accuracy: Nonetheless, the accuracy of sentiment classification should be at least 85%.
- Reliability: The system should collect data every day under any circumstances.
- Response Time: The results should be loaded within five seconds on the dashboard.
- Scalability: Should support and analyze about 50,000 reviews every day.
- Security: No private data on users shall be stored, and reviews shall be anonymized.
- Explainability: The sentiment will highlight keywords

4. Conclusion

- Sentiment Analysis: An AI technique that classifies text as positive, negative, or neutral.
- NLP: AI methods concerning the handling and analysis of human language.
- Knowledge Base: Structured storage of cleaned reviews and sentiment metadata.